cytoscape basics

- install
- creating networks
- working with networks
- visual effects

Install cytoscape

- cytoscape is a software based in java
- install first the compatible java11 environment https://adoptopenjdk.net/ (https://adoptopenjdk.net/)
- test you environment for all needed components: https://cytoscape.org/troubleshooting.html (https://cytoscape.org/troubleshooting.html)
- download cytoscape for your operating system: https://cytoscape.org/download.html (https://cytoscape.org/download.html)
- choose to create an icon on your desktop, so you can start it

First steps

Before we start: what are your expectations from a visualization tool?

- interactivity for changing networks
- easy to use menus
- graphical and tabular views, possibly parallel
- compatibility with the main data formats
- extensions and batch processing option

The cytoscape Interface

The most important points:

- File menu: for general tasks
- Edit menu: for modifying networks
- Tools menu: for installing new modules
- View menu: for changing the visible panes

The first network

- Select the File menu and create a new, empty network!
- Click with you mouse on the middle of the left pane: node is created
- Add several new nodes with mouse clicks
- Connect the nodes with edges by selecting one node, and draw a line to another node
- You see other features in the left and right bottom pane
    - left pane: general characteristics,
    - bottom pane: table with node properties

Import/output of networks

- Select the File menu and click "open"
- Choose the word association network from the file list!
- Select "Save": the network is updated
- Select "Save as": a new ".cys" file is created

File formats: cys

- Open the new .cys file with a file archiver tool!
- You see several xml files zipped and packaged
- Extract one gxmml file, and open it with a text editor
- There are several IDs, node/edge properties are xml tags

Identifyers if network objects

- Create a network of 2 nodes and connect them!
- Clone if with File -> new network -> clone steps
- Save the network in a .cys file:
    - All networks are saved (with session related infos)
    - Compare the IDs of the original and the cloned network!

Simpler file formats

- Create a two-column list of random words with a text editor
- Add two columns with numbers to each row
- Import this file from the "Import network" menu
- Specify the source and target columns and columns for edge properties
- Import this file from the "Import node properties", select the source column as "key"

Interacting with other tools

- The xgmml and cys formats are for complex formatting
- The txt format is for interacting with other, general tools

Usual work flow:

- Work with a batch processing tool
- Visualize and analyze small parts with cytoscape

Modifying a network

- Click on a node, and drag it with your mouse
- Click near to a node, and select node and edge with a selection box
    - Click on a selected node and move it around
- Click on an unselected node or an empty place: deselect all
- Select some nodes with a box, and go to "Select" menu and "invert" your selection

Modifying node properties

- Select a node with a mouse click
- Go to the bottom pane, select the "nodes" tab
- Change the visible properties with the first icon
- Add a new property with the "+" icon
- Click on a cell in the column of the new property and add a new value

Change visual properties

- Go to the "Style" tab on the general pane
- Select "node" tab at the bottom
- The most useful properties:
  - node color
  - node size
  - node shape

Visual property types: float

- Select node size
- Select a node attribute of float numbers
- Select "mapping type"
- Select minimal and maximal values for a linear map
- Add additional mapping points to nonlinear mapping
    - This is a piecewise linear mapping

Visual property: discrete

- Select node shape
- Select a node attribute with string values
- Select mapping type: discrete
- From the list of possible values, select corresponding shape

Visual property: generated

- Select node color
- Select a node attribute with integer values
- Select mapping type: discrete
- With right mouse click: select automated filling
- Select an appropriate color scale

Visual property: exceptions

- Select a node with right mouse click
- From the drop down menu select "Visual bypass"
- Select the visual property (e.g. color)
- Define the exceptional value for the property

Searching in networks

- Go to the top right search field
- Type at least 3 characters of the name of an existing node
- Type at least 3 characters that are common in several nodes
- Select a node and from "Select menu" choose the "first neighbours"
    - repeat selecting the first neighbours: count how many times you need until all nodes get selected

Complex search

- Go to the Filter tab on the left pane (network global properties)
- Add a new rule with selecting a node property
- Add a new rule with selecting another node property
- Adjust the filter by selecting a range
- Create a complex rule with connecting elementary rules with "and", "or" and "not"

Select from tabular view

- Go to the tabular pane of node attributes
- Select column with values of float numbers
- Click on the header to order the values
- Select with shift + cursor_down
- With right click select nodes on the network view